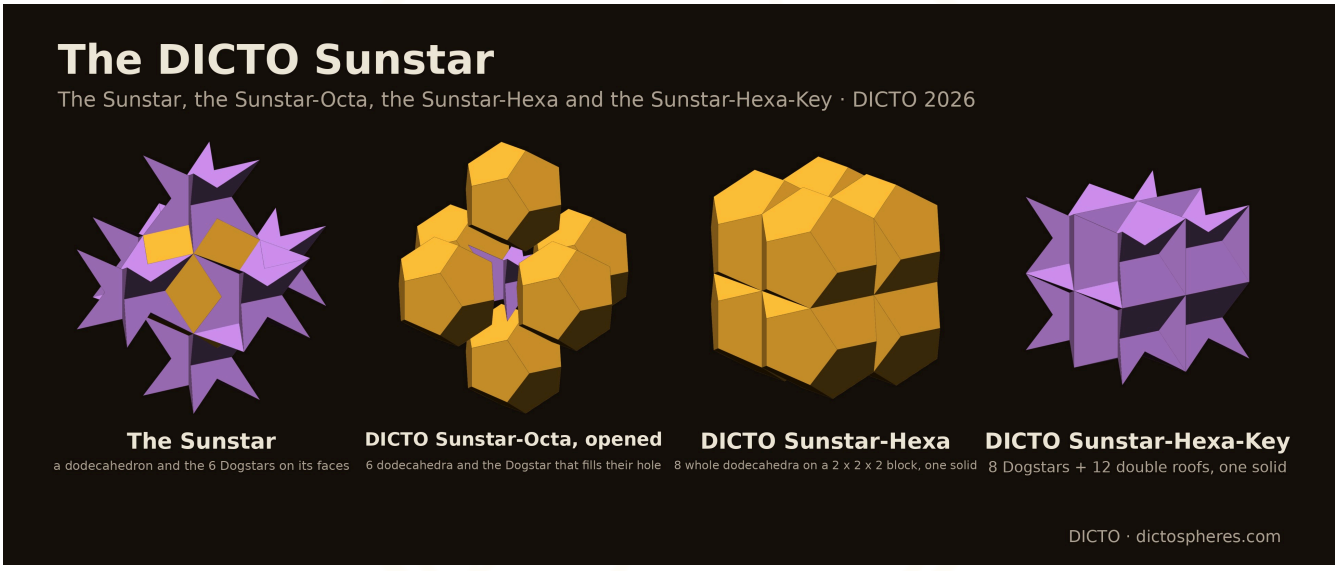
The DICTO Sunstar

Regular dodecahedra and Dogstars fill space: the Dogstar as an exact stellation, the Star Chain Reaction, the DICTO Sunstar-Tetra, Sunstar-Octa and Sunstar-Rhombi, and the DICTO Sunstar-Hexa and Sunstar-Hexa-Key, all exact; the twin of the DICTO Jewel

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Version 1 · 20 October 2026 · DICTO DOI: 10.5281/zenodo.23497702 · Licence CC BY-SA 4.0 · Scripts, exact checks and data: https://github.com/DICTOR-Master/kaleidohedra/tree/dc78106/studies/2026-10-20-sunstar (research branch, commit dc78106; also attached to this record) · Interactive: dictospheres.com → Kaleidohedra → DICTO → Sunstar Lattice

The names Dogstar, Sunstar, Sunstar Lattice, Star Chain Reaction, Sunstar-Tetra, Sunstar-Octa, Sunstar-Rhombi, Sunstar-Hexa and Sunstar-Hexa-Key, the clusters and the Hexa twin are DICTO's (findings #12 and #16 on dictospheres.com); computation, exact proofs and drafting with Claude (Anthropic). The Dogstar as a solid was illustrated by John Ounsted, by Grünbaum and Shephard and by Stewart, as listed by George W. Hart, who gave it as stellation 8 of the dodecahedron and noted its tiling with regular dodecahedra (1996) [6]; the tiling is also in Hans Walser [7] and on Polyhedra-World [8].



Abstract. Regular dodecahedra on alternate cubes of a cubic lattice (Euclid's dodecahedron: a cube with a roof on each face) are the densest lattice packing of the dodecahedron [9]; the hole left in each other cube is a star, the **Dogstar**, and dodecahedra and Dogstars fill space. The Dogstar and this tiling are Hart's [6]. We prove them exactly in $\mathbb{Q}(\sqrt{5})$: the Dogstar is the cube minus the six roofs that reach into it, of volume exactly $\varphi/2$ at dodecahedron edge 1, and, as a stellation of the dodecahedron (Hart's), we identify it exactly: the dodecahedron $1/\varphi^3$ the size of the lattice's, with all 12 pyramids, 24 of the 30 wedges and 8 of the 20 spikes; it lies inside its own cube's stella octangula. The **Star Chain Reaction**: the great stellated dodecahedron of that core has the cell dodecahedron's 20 corners, and a whole **Sunstar** (a dodecahedron with the Dogstars on its six faces) $1/\varphi^3$ the size fits inside it exactly, with no room to grow, and so on for ever, ratio φ^3. DICTO's clusters become single solids: the **DICTO Sunstar-Tetra** (4 dodecahedra), **Sunstar-Octa** (6 dodecahedra and the Dogstar that exactly fills their hole: a Sunstar turned inside out) and **Sunstar-Rhombi** (8 dodecahedra round a sealed Dogstar). Neighbouring dodecahedra touch on exactly one thick Penrose rhombus, the window of the DICTO Jewel [2]; the clusters sit on the Jewel clusters' cells, so their tilings

with loose Dogstars (4, 12 and 25 lattices; 2, 1 and 9 up to symmetry) and the octet pattern are those of the Jewel. Eight whole dodecahedra on a 2 × 2 × 2 block make one solid, the **DICTO Sunstar-Hexa** ((9 + √5)/8 of the block); the space between such blocks is one solid, the **DICTO Sunstar-Hexa-Key** ((7 − √5)/8: 8 Dogstars and 12 double roofs); they fill space in an irrational ratio, (17 + 4√5)/11, where the Jewel's Hexa and Hexa-Key are 5 : 3. The two lattices have the same cells and roofs: a Jewel is a dodecahedron without its six roof tips, and a stella octangula is a Dogstar with them. All claims are exact except where marked.

1. The Dogstar and the Sunstar Lattice (exact)

Frame. As in the EKP cell [1] and the Jewel [2]: cells are cubes centred on a cubic lattice; a cell is *even* or *odd* by the parity of its index sum. All values are at the scale of [2], where a 2 × 2 × 2 block of cells (the Sunstar-Hexa's cube, §4) has edge 1, so a cell has edge 1/2; the scripts use cells of edge 2, where every volume is 64 times larger. Euclid builds the regular dodecahedron on a cube by setting a roof on each face [4]; each roof is (1 + √5)/192 and the dodecahedron (5 + √5)/32.

The Dogstar. Put a dodecahedron on every even cell. Each odd cell is entered by six roofs, one from each face-neighbour; they lie in the cell and meet each other only in faces (exact: pairwise intersection volume 0). What is left is the **Dogstar** (DICTO's name), of volume 1/8 − 6 × (1 + √5)/192 = **(3 − √5)/32**, exactly φ/2 at dodecahedron edge 1. A dodecahedron is its cell and six roofs, each lying in a face-neighbour cell (exact); so only these six roofs enter an odd cell, and two dodecahedra can meet only there, where they are roofs that meet in faces alone. Hence **dodecahedra and Dogstars fill space**; the dodecahedra take (5 + √5)/8 of it, the density of Betke and Henk's densest lattice packing [9, 10]. This solid is Hart's stellation 8 of the dodecahedron, earlier illustrated by Ounsted, by Grünbaum and Shephard and by Stewart (as listed in [6]); the tiling is Hart's [6] (also [7, 8]). What follows is new.

Which stellation, exactly. Take the regular dodecahedron $1/\varphi^3$ the size of the lattice's, centred in the odd cell and turned the same way (the *core*). Its 12 face planes cut space into 63 bounded cells: the core, 12 pyramids, 30 wedges and 20 spikes. Every one lies wholly inside the Dogstar or wholly outside it, and the Dogstar is exactly **the core, all 12 pyramids, 24 of the 30 wedges and 8 of the 20 spikes**, their volumes adding up to (3 − √5)/32. The 6 wedges left out lie on the cube's axes; the tips of the 8 spikes kept are the cube's 8 corners. The Dogstar also lies wholly inside its own cube's stella octangula (exact), so Dogstars can sit in every cell, eight tips meeting at each cube corner.

The Sunstar (DICTO's name): a dodecahedron with the Dogstars of its six face-neighbour cells, (23 − 5√5)/32.

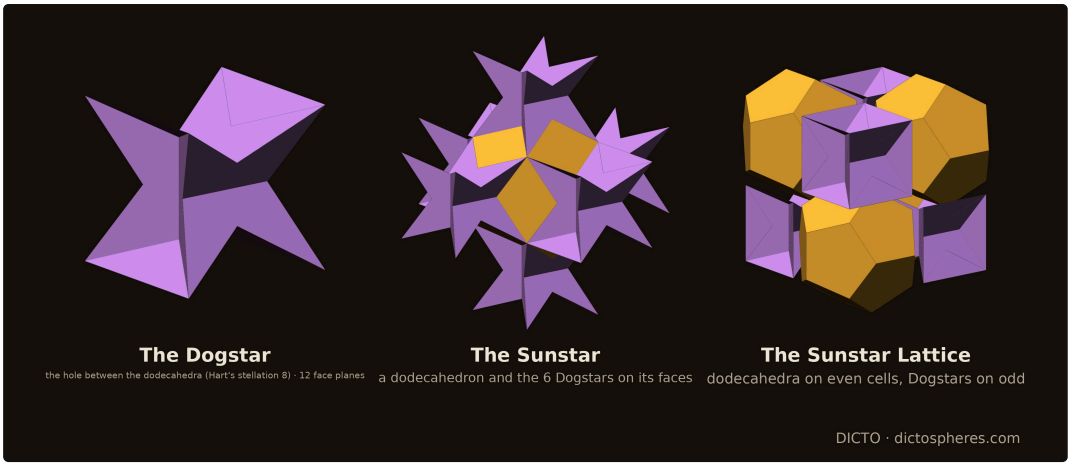


Figure 1. The Dogstar; the Sunstar (gold dodecahedron, purple Dogstars); a 2 × 2 × 2 block of the Sunstar Lattice, drawn slightly apart.

2. The Star Chain Reaction (exact)

The great stellated dodecahedron [5] of the core (all 63 cells) has exactly the 20 corners of the lattice's dodecahedron, φ^3 times the core (classical [15]; verified exactly here): it is the great star of the EKP cell [1], and the Dogstar is this star cut back to its cube. Inside it sits a **whole Sunstar $1/\varphi^3$ the size**, centred with it and turned the same way: the core itself with the six Dogstars of the $1/\varphi^3$ lattice round it. Each of those Dogstars lies wholly inside the great star (exact: its volume met in full), and 24 of their spike tips (the four outermost of each) lie on the great star's surface in such a way that any enlargement about the centre takes them outside: **the fit is tight, at exactly $1/\varphi^3$**. For this the great star must be star-shaped about its centre, which it is: a ray from the centre crosses each of the 12 planes once, so the planes it has crossed only accumulate, and every subset of a bounded cell's outer planes is again a bounded cell's (checked for all 63), so a tip once outside stays outside. The core holds the next great star, which holds the next Sunstar $1/\varphi^6$, and so on: dodecahedron $\supset$ great star $\supset$ Sunstar$(1/\varphi^3)$ $\supset$ great star$(1/\varphi^3)$ $\supset$ … (DICTO's Star Chain Reaction, finding #12). The ratio φ^3 is the inflation multiplier characteristic of the primitive (P-type) icosahedral tiling [16].

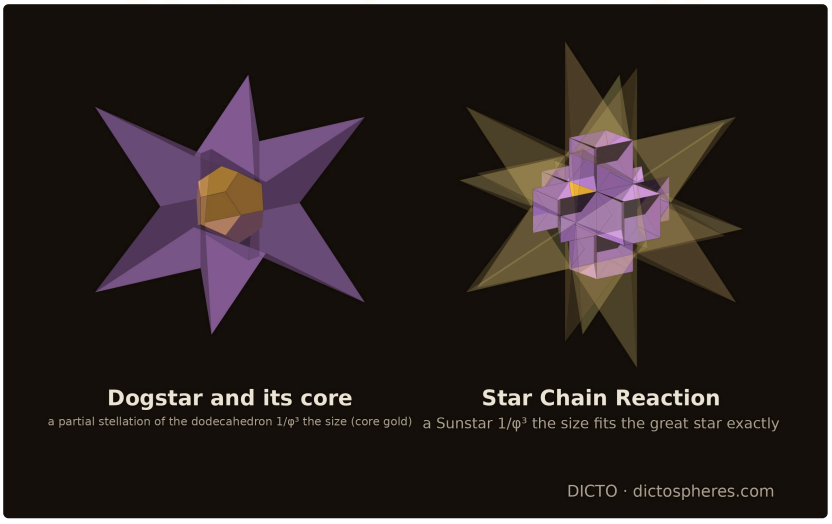


Figure 2. Left: the Dogstar, see-through, round its core (gold), the dodecahedron $1/\varphi^3$ the size. Right: the Star Chain Reaction, a Sunstar $1/\varphi^3$ the size inside the great star (pale gold, see-through).

3. The Sunstar-Tetra, Sunstar-Octa and Sunstar-Rhombi (exact)

Contacts. Dodecahedra on neighbouring even cells (one step along a face diagonal) do not overlap and touch on **exactly one thick Penrose rhombus**, found by clipping their coplanar faces exactly: two cube corners, a ridge point and a point inside the face, with 108° at the ridge point (exact). Since the two solids are convex and do not overlap, their whole contact lies in one common face plane, so this rhombus is all of it. It is the window of the DICTO Jewel [2]; what is left of the pentagon is Walser's semi-regular pentagon, a regular pentagon with a rhombus cut away [7]. Dodecahedra two cells apart along an axis do not touch.

The DICTO Sunstar-Tetra (DICTO's tetrahedral cluster, finding #16, as one solid): four dodecahedra round a cell corner, each pair meeting on one rhombus (6 in all), all four sharing that corner, where four Dogstar tips meet; volume $(5 + \sqrt{5})/8$.

The DICTO Sunstar-Octa (DICTO's octahedral cluster): the six dodecahedra round an odd cell, meeting on 12 rhombi, and that cell's Dogstar. Their six roofs are exactly the roofs that make up the odd cell with the Dogstar (§1), so **the Dogstar fills their hole exactly**; the 12 contact rhombi lie across the odd cell's 12 edges, sealing it. A Sunstar is a dodecahedron with its six Dogstars; the Sunstar-Octa is a Dogstar with its six dodecahedra, the Sunstar turned inside out. Volume $(33 + 5\sqrt{5})/32$.

The DICTO Sunstar-Rhombi: eight dodecahedra on the corners of the primitive (rhombohedral) cell of the face-centred cubic lattice, meeting on **18 rhombi** (12 along its edges, 6 along the short face diagonals), with the Dogstar of the odd cell at its centre sealed inside, as in the Sunstar-Octa; volume $(43 + 7\sqrt{5})/32$.

How they fill space. The fillers are always the dodecahedra and the Dogstars. Each cluster sits on the same cells as the Jewel-Tetra, Jewel-Octa and Jewel-Rhombi [2], so each fills space with copies of itself and *loose* Dogstars (Dogstars inside no piece) on exactly the lattices found there: 4, 12 and 25 lattices of translations (2, 1 and 9 up to the piece's symmetry), with 4, 5 and 7 loose Dogstars per piece. Shared, they form the octet pattern (8 Sunstar-Tetras round each Sunstar-Octa) and the pyrochlore network of Sunstar-Tetras (DICTO's Kagome network); without overlap at most one Dogstar in six can be inside a Sunstar-Octa.

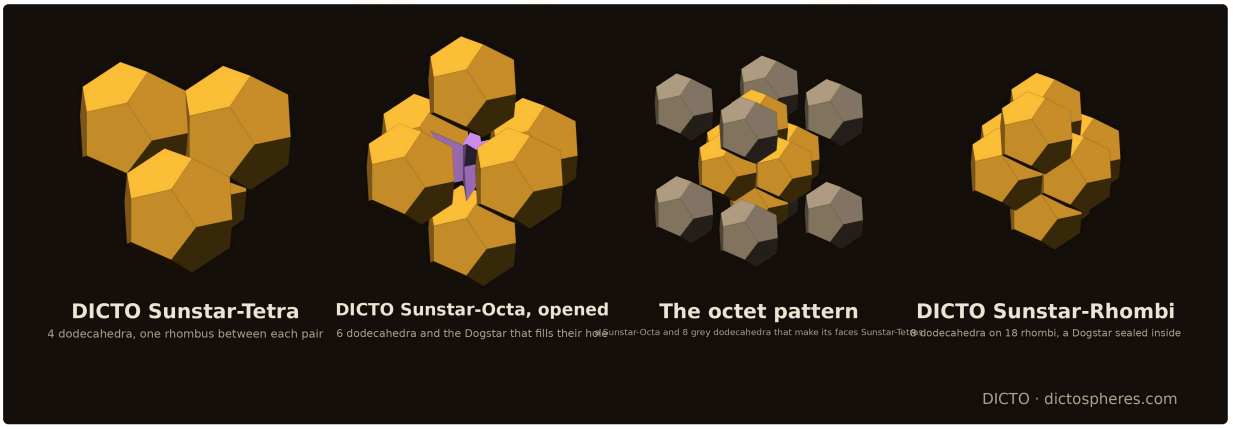


Figure 3. The DICTO Sunstar-Tetra; the DICTO Sunstar-Octa, its dodecahedra drawn out to show the Dogstar (purple) that fills its hole; the octet pattern, a Sunstar-Octa (gold) and the 8 dodecahedra (grey) that complete its faces to Sunstar-Tetras, drawn out; the DICTO Sunstar-Rhombi.

4. The DICTO Sunstar-Hexa and Sunstar-Hexa-Key (exact)

The twins of DICTO's Hexa and Hexa-Key [13, 2]. Put a whole dodecahedron on each of the eight cells of a 2 × 2 × 2 block, even and odd cells alike. Inside the block they pass through each other; out of it reach 24 roofs, pairwise without overlap. As one solid, the **DICTO Sunstar-Hexa** is the block with 24 roofs: $1 + 24 \times (1 + \sqrt{5})/192 = \mathbf{(9 + \sqrt{5})/8}$. Put Sunstar-Hexas on alternate blocks, rock-salt fashion; each other block is entered by 24 roofs from its six neighbours, three into each of its cells, and what is left is one solid, the **DICTO Sunstar-Hexa-Key**: $1 - 24 \times (1 + \sqrt{5})/192 = \mathbf{(7 - \sqrt{5})/8}$, eight Dogstars joined by twelve double roofs: each Key cell is its Dogstar and the three roofs on its inner faces, and across each of the 12 inner faces the two roofs share their base (exact). Its recesses are the Sunstar-Hexas' roofs, so they fill space, in the ratio $\mathbf{(17 + 4\sqrt{5})/11 \approx 2.358}$, irrational, where the Jewel's Hexa and Hexa-Key are 5 : 3. Each Sunstar-Hexa meets each of its 12 neighbours on exactly two rhombi (24 in all) and no two overlap (checked in full; contacts are counted over cells one face diagonal apart, since by §1 no other pair of dodecahedra can share area).

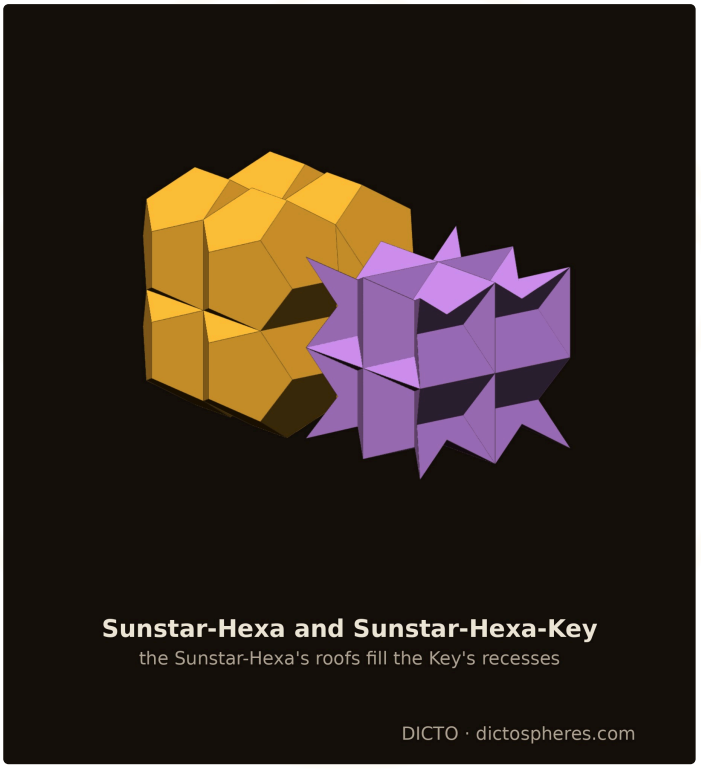


Figure 4. A Sunstar-Hexa (gold) and the Sunstar-Hexa-Key (purple) next to it, drawn out a little: the Sunstar-Hexa's roofs go into the Key's recesses.

5. The Sunstar and the Jewel

The Sunstar Lattice and the Stella–Jewel Lattice [2] have the same cells and the same roofs; they differ only in who owns the tip of each roof, the part $(\sqrt{5} - 1)/192$ that lies inside the neighbouring cell's stella octangula. A dodecahedron without its six tips is the DICTO Jewel; a Dogstar with them is the stella octangula:

	even cell	odd cell	clusters	Hexa : Hexa-Key
Sunstar Lattice	dodecahedron $(5 + \sqrt{5})/32$	Dogstar $(3 - \sqrt{5})/32$	Sunstar-Tetra, -Octa, -Rhombi	$(17 + 4\sqrt{5})/11$
Stella–Jewel Lattice [2]	Jewel 3/16	stella octangula 1/16	Jewel-Tetra, -Octa, -Rhombi	5 : 3

So the two families share their contacts (the Penrose windows), their clusters and every tiling, and the clusters look alike; the dodecahedra show whole pentagons where the Jewels show windows and triangles.

6. Prior art and what is claimed

- **Known and credited.** The Dogstar as a solid and its tiling with regular dodecahedra: Hart's stellation 8 (1996) [6], Walser's semi-regular dodecahedron [7], Polyhedra-World [8]. The densest lattice packing of the dodecahedron, density $(5 + \sqrt{5})/8$: Betke and Henk [9] (also [10]). Euclid's roofs [4], Kepler's great stellated dodecahedron [5] and its corners on the dodecahedron φ^3 times its core [15], the octet truss [14], the inflation multiplier of icosahedral tilings [16].
- **Already published by DICTO.** The Sunstar Lattice, the Star Chain Reaction (#12) and the Sunstar clusters and octet network (#16) were first released with the Kaleidohedra app [11]; the Sunstar hull clusters and their Kagome networks are in [3]; the Jewel family and its tilings in [2]; the Hexa and Hexa-Key in [13].
- **Claimed here:** the exact proofs of the Dogstar's volume and of the tiling; as a stellation of the dodecahedron the Dogstar is Hart's [6], and new here is its exact identification on the core $1/\varphi^3$ of the lattice's dodecahedron (which cells) and of the Dogstar inside its stella octangula; the exact Star Chain Reaction, including the tight fit at exactly $1/\varphi^3$; the contacts as exact Penrose rhombi; the DICTO Sunstar-Tetra, Sunstar-Octa and Sunstar-Rhombi as single solids, with the Dogstar filling the hole exactly, and their tilings; the DICTO Sunstar-Hexa and Sunstar-Hexa-Key, their filling and irrational ratio; and the Sunstar–Jewel correspondence.

In the sources checked (the full text of the library sources listed in the appendix, and web searches on 8–9 and 20 October 2026) we did not find the Star Chain Reaction, the Sunstar clusters as solids, the Sunstar-Hexa or the Sunstar-Hexa-Key.

7. Further work

- The Sunstar-Hexa and Sunstar-Hexa-Key in the Kaleidohedra app and Polyhedraverse, with a helper that shows where the next piece fits.
- The Star Chain Reaction as a self-similar (φ^3) packing: what fills the space between the great star and the Sunstar inside it.

Data and reproducibility

Exact ($\mathbb{Q}(\sqrt{5})$): sunstar_exact.py (dodecahedron and roofs, the Dogstar and the tiling, the 63 stellation cells of the core, the Star Chain Reaction and its tight fit; log sunstar-exact.log, 12 PASS, with the great star star-shaped about its centre) and sunstar_clusters.py (contacts by exact clipping, the three clusters, the Sunstar-Hexa and Sunstar-Hexa-Key, the Dogstar in its stella octangula, the volumes at the paper's unit; log sunstar-clusters.log, 16 PASS, with the contact angle and the Key's double roofs); both run with the exact tools of [12] (python3 <script> studies/2026-10-18-ocd-octet/scripts). The tilings of the clusters are those of the Jewel clusters, checked in [2] (cluster_exact.py). The face order in sunstar_clusters.py's faces() is a float sort, which does not affect any result. Figures: sunstar_figs.py (float copies of the exact pieces), shots.mjs (2× pixel ratio), sunstar_compose.py.

References

1. DICTO, The Euclid–Kepler–Pacioli Cell, DICTO DOI: 10.5281/zenodo.23284106.
2. DICTO, The DICTO Jewel, DICTO DOI: 10.5281/zenodo.23496353.
3. DICTO, Kagome Hulls, DICTO DOI: 10.5281/zenodo.23376317.
4. Euclid, The Thirteen Books of the Elements, tr. T. L. Heath, vol. III (Books X–XIII), Cambridge (1908), Book XIII, Proposition 17.
5. J. Kepler, Harmonices mundi libri V (1619), Book II.
6. G. W. Hart, Tetrahedral stellations of the dodecahedron (1996), https://www.georgehart.com/virtual-polyhedra/stellations-dodecahedron-tetrahedral.html (stellation 8; it lists J. Ounsted, An unfamiliar dodecahedron; B. Grünbaum and G. C. Shephard, Duality of polyhedra; B. M. Stewart, Adventures among the Toroids, p. 255).
7. H. Walser, Semi-regular figures: between beauty and regularity (talk script, 27 June 2017), https://walser-h-m.ch/hans/Vortraege/20170627/script.htm.
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11. DICTO, Kaleidohedra by DICTO, DICTO DOI: 10.5281/zenodo.23173809; releases 10.5281/zenodo.23256623 (#16) and 10.5281/zenodo.23257393.
12. DICTO, The DICTO Ogdoecontahedron on the Octet Truss, DICTO DOI: 10.5281/zenodo.23457016 (the exact tools q5.py and exactpoly.py).
13. DICTO, The DICTO Hexa, DICTO DOI: 10.5281/zenodo.23376320.
14. R. B. Fuller, Octet truss, US Patent 2,986,241 (1961).
15. H. S. M. Coxeter, Regular Polytopes, 3rd ed., Dover, New York (1973).
16. M. Baake, A guide to mathematical quasicrystals, arXiv:math-ph/9901014 (1999).

Appendix. The reference library searched

The full text of the sources below in DICTO's reference library was searched on 20 October 2026 for: stellation, great stellated, Betke, Hart, Walser, dodecahedra and inflation. None contains the results of this paper. The relevant hits are Betke and Henk's densest lattice packings [9] and the packing studies citing it (Torquato and Jiao; de Graaf, van Roij and Dijkstra, whose dense packings of the great stellated dodecahedron interpenetrate and are not this structure), Messer's stellations of the rhombic triacontahedron, and the icosahedral tiling and quasicrystal papers on inflation (Baake, Grimm, Gähler; Koca et al.; Dietl and Eschenburg). Thirteen further items are page images without a text layer (the patents of Fuller, Snelson, Pearce, Baer, Hogan, Bergman and Lalvani, and Fedorov 1885).

Web searches (20 October 2026): a dodecahedron stellation filling space with regular dodecahedra (Hart [6]); clusters of regular dodecahedra in the densest lattice packing, octahedral and tetrahedral clusters round the holes (nothing on clusters).

Author(s)	Year	Title
Luca Pacioli, Leonardo da Vinci (ill.)	1509	Divina proportione
Albrecht Dürer	1525	Underweysung der Messung mit dem Zirckel und Richtscheyt
Wenzel Jamnitzer, Jost Amman (engr.)	1568	Perspectiva corporum regularium
Johannes Kepler	1596	Prodromus dissertationum cosmographicarum continens Mysterium cosmographicum
Johannes Kepler	1619	Harmonices mundi libri V
Euclid, T. L. Heath (tr.)	1908	The Thirteen Books of Euclid's Elements, Vol. III (Books X-XIII)
M. Goldberg	1937	A class of multi-symmetric polyhedra
Haresh Lalvani	1990	Transformational part-count in layered octahedral-tetrahedral truss configurations (NASA TM-102635)
P. W. Messer	1995	Stellations of the rhombic triacontahedron and beyond

M. Baake, F. Gähler	1998	Symmetry structure of the Elser-Sloane quasicrystal
T. C. Hales	1998	An overview of the Kepler conjecture
M. Baake	1999	A guide to mathematical quasicrystals
O. Ogievetsky, Z. Papadopolos	1999	On Quasiperiodic Space Tilings, Inflation and Dehn Invariants
U. Betke, M. Henk	1999	Densest lattice packings of 3-polytopes
Z. Papadopolos, O. Ogievetsky	1999	On Inflation Rules for Mosseri-Sadoc Tilings
Z. Papadopolos, O. Ogievetsky	1999	On quasiperiodic space tilings, inflation and Dehn invariants
R. V. Moody	2000	Model sets: a survey
M. Senechal	2004	Donald and the golden rhombohedra
S. Kabai	2007	54-Faced Space-Filling Polyhedron (Wolfram Demonstration; description and screenshot)
S. Kabai	2007	Golden Octet Truss (Wolfram Demonstration; description and screenshot)
Wikipedia contributors (2007 mirror)	2007	Kyushindo (2007 version)
S. Kabai	2008	A Rhombic Dodecahedron Made of Rhombic Triacontahedra (Wolfram Demonstration; description and screenshot)
S. Kabai	2008	Relating a Rhombic Triacontahedron and a Rhombic Dodecahedron (Wolfram Demonstration; description and screenshot)
M. Koca, M. Al-Ajmi, N. O. Koca	2009	Grand antiprism and quaternions
M. Koca, M. Al-Ajmi, N. O. Koca	2009	Quaternionic representation of snub 24-cell and its dual polytope derived from E8 root system
S. Torquato, Y. Jiao	2009	Dense packings of polyhedra: Platonic and Archimedean solids
B. Grünbaum	2010	The Bilinski dodecahedron, and assorted parallelohedra, zonohedra, monohedra, isozonohedra and otherhedra
P. Kramer	2010	Gateways towards quasicrystals
A. M. Cutler, E. Schulte	2011	Regular polyhedra of index two, I
D. Frettlöh	2011	Icosahedral tilings in R3: the ABCK tilings (draft chapter)
J. Koskinen	2011	Unusual tilings and transformations
J. de Graaf, R. van Roij, M. Dijkstra	2011	Dense regular packings of irregular non-convex particles
M. Koca, N. O. Koca, M. Al-Ajmi	2011	Branching of the W(H4) polytopes and their dual polytopes under the Coxeter groups W(A4) and W(H3) represented by quaternions
M. Koca, N. O. Koca, M. Al-Barwani	2011	Snub 24-cell derived from the Coxeter-Weyl group W(D4)
T. Etzion	2011	Product constructions for perfect Lee codes
A. Garber, A. Gavrilyuk, A. Magazinov	2012	The Voronoi conjecture for parallelohedra with simply connected delta-surface
J. C. Lagarias, C. Zong	2012	Mysteries in packing regular tetrahedra
M. Baake, U. Grimm	2012	Mathematical diffraction of aperiodic structures
M. Koca, N. O. Koca, R. Koc	2012	Affine Coxeter group Wa(A4), quaternions, and decagonal quasicrystals
P. F. Damasceno, M. Engel, S. C. Glotzer	2012	Crystalline assemblies and densest packings of a family of truncated tetrahedra and the role of directional entropic forces
R. Gabbrielli, Y. Jiao, S. Torquato	2012	Families of tessellations of space by elementary polyhedra via retessellations of face-centered-cubic and related tilings
S. Torquato, Y. Jiao	2012	Organizing principles for dense packings of nonspherical hard particles: not all shapes are created equal
M. Kléman	2013	Defects in quasicrystals, revisited: I-flips, approximants, phason defects
N. Fujita, H. Takano, A. Yamamoto, A.-P. Tsai	2013	Cluster packing geometry for Al-based F-type icosahedral alloys
P.-P. Dechant, J. Wardman, T. Keef, R. Twarock	2014	Viruses and fullerenes: symmetry as a common thread?
A. Campello, G. C. Jorge, J. E. Strapasson, S. I. R. Costa	2015	Perfect codes in the lp metric
K. Deguchi, M. Nakayama, S. Matsukawa, K. Imura et al.	2015	Crystal structure of superconducting 1/1 cubic Au-Ge-Yb approximant with Tsai-type cluster
M. Baake, D. Damanik, U. Grimm	2015	What is aperiodic order?
M. Firsching	2015	Computing maximal copies of polytopes contained in a polytope
N. O. Koca, M. Koca, A. Al-Mukhaini, A. Al-Qanobi	2015	Quaternionic representations of the pyritohedral group, related polyhedra and lattices
R. Twarock, M. Valiunas, E. Zappa	2015	Orbits of crystallographic embedding of non-crystallographic groups and applications to virology
S. Kabai	2015	Cluster of 14 Rhombic Triacontahedra (Wolfram Demonstration; description and screenshot)
A. T. Cadotte, J. Dshemuchadse, P. F. Damasceno, R. S. Newman, S. C. Glotzer	2016	Self-assembly of a space-tessellating structure in the binary system of hard tetrahedra and octahedra
C. Hann, J. E. S. Socolar, P. J. Steinhardt	2016	Local growth of icosahedral quasicrystalline tilings

R. M. K. Dietl, J.-H. Eschenburg	2016	The icosahedral quasiperiodic tiling and its self-similarity
A. Stergiou	2018	Bootstrapping hypercubic and hypertetrahedral theories in three dimensions
A. Garber	2019	Voronoi conjecture for five-dimensional parallelohedra
D. Poland, S. Rychkov, A. Vichi	2019	The conformal bootstrap: theory, numerical techniques, and applications
R. Twarock, A. Luque	2019	Structural puzzles in virology solved with an overarching icosahedral design principle
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